

Experiment-3

Aim: To install a **virtualization platform** (VirtualBox or VMware Workstation) on a **Windows 8 or later system** and run **different operating systems** (Linux / Windows) as **virtual machines**.

Description:

1. What is Virtualization?

Virtualization allows a **single physical computer** to run **multiple operating systems simultaneously** by creating **virtual machines (VMs)**.

- **Host OS** → Main operating system (Windows 8 or above)
- **Guest OS** → OS running inside the virtual machine

2. Virtualization Software

Software	Type
VirtualBox	Type-2 Hypervisor
VMware Workstation	Type-2 Hypervisor

→ Type-2 hypervisors run **on top of an existing OS**.

3. System Requirements

Hardware Requirements

- 64-bit processor with **Virtualization support**
 - Intel VT-x or AMD-V
- Minimum **4 GB RAM** (8 GB recommended)
- At least **30 GB free disk space**

Software Requirements

- Host OS: **Windows 8 / 8.1 / 10 / 11**
- Virtualization software:
 - Oracle VirtualBox **OR**
 - VMware Workstation
- Guest OS ISO files:
 - Ubuntu / Fedora / Debian (Linux)
 - Windows 10 / 11 ISO (optional)

4. Pre-Installation Setup

Enable Virtualization in BIOS

1. Restart the computer
2. Enter BIOS/UEFI (F2, F10, DEL, or ESC)
3. Enable:
 - **Intel Virtualization Technology** OR
 - **SVM / AMD-V**
4. Save and exit

✓ Required for running 64-bit VMs

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Step 2: Download Required Software

- VirtualBox: <https://www.virtualbox.org>
- VMware Workstation: <https://www.vmware.com>
- Linux ISO (example):
 - Ubuntu: <https://ubuntu.com/download>

1. Installing VirtualBox (Example)

Step 3: Install VirtualBox

1. Run the VirtualBox installer
2. Click **Next**
3. Select installation options
4. Allow network interface installation
5. Click **Install**
6. Finish installation

✓VirtualBox installed successfully

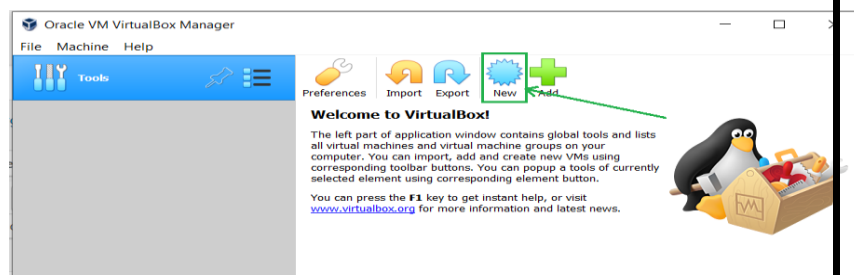
2. Creating a Virtual Machine (Linux Example)

Steps to Install Ubuntu on VirtualBox

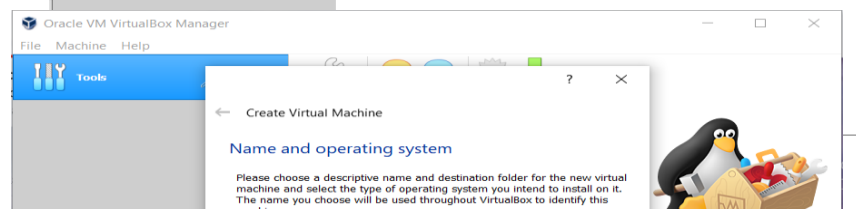
Step 1: Before we begin with the installation process, we need to **Download ISO for Ubuntu**. For that, all the versions of Ubuntu are available on the [Official Site](#).

Step 2: Open [VirtualBox](#) and click on the **New** button

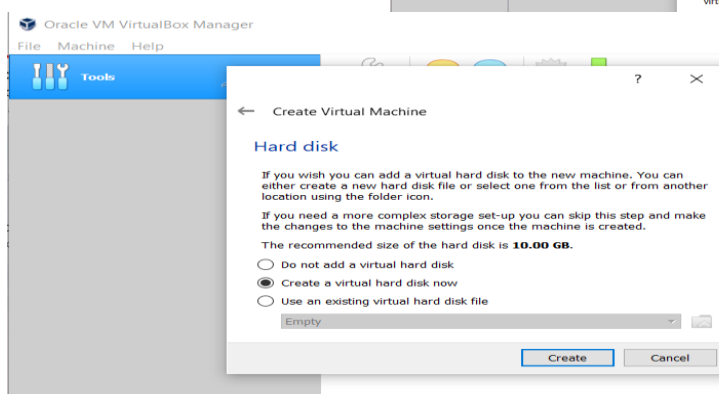
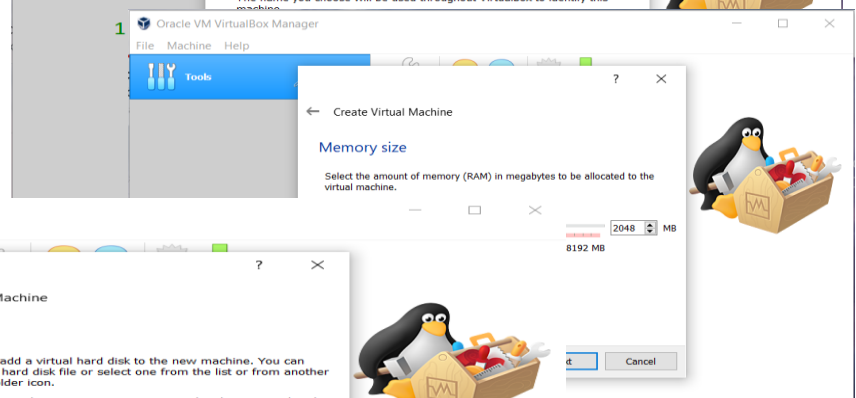
Step 3: Give a **Name** to your **Virtual Machine** and select the **Location** for it to install.



Step 4: Assign **RAM Size** to your Virtual Machine.

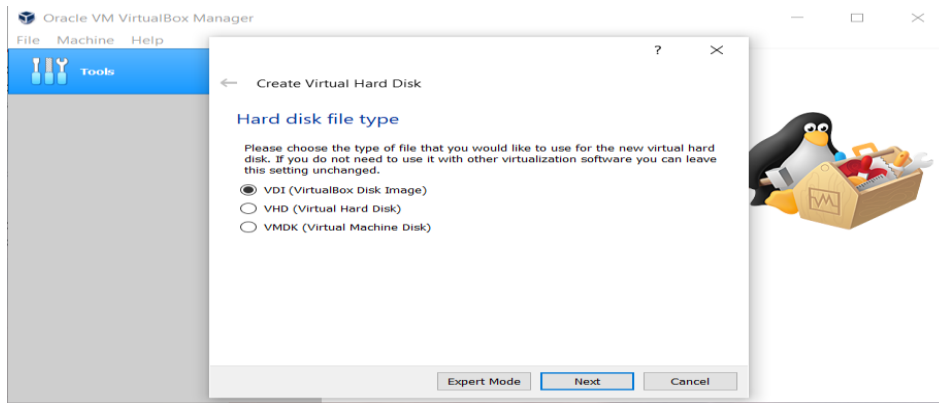


Step 5: Create a **Virtual Hard Disk** for the machine to store files.

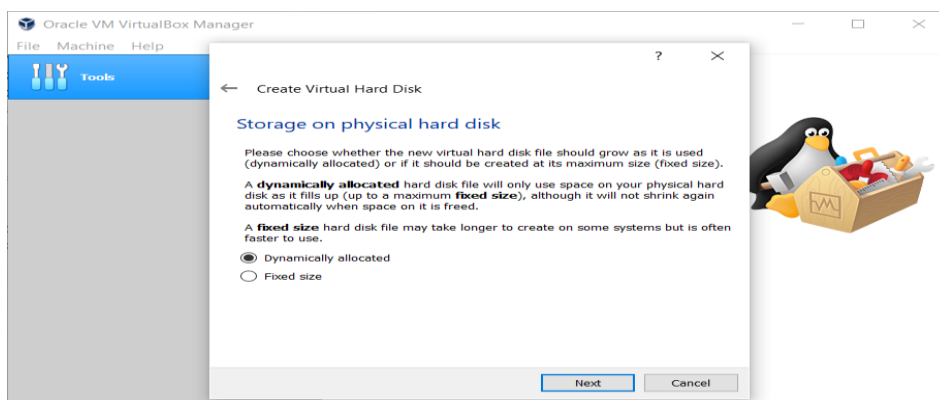


Step 6: Select the type of Hard disk.

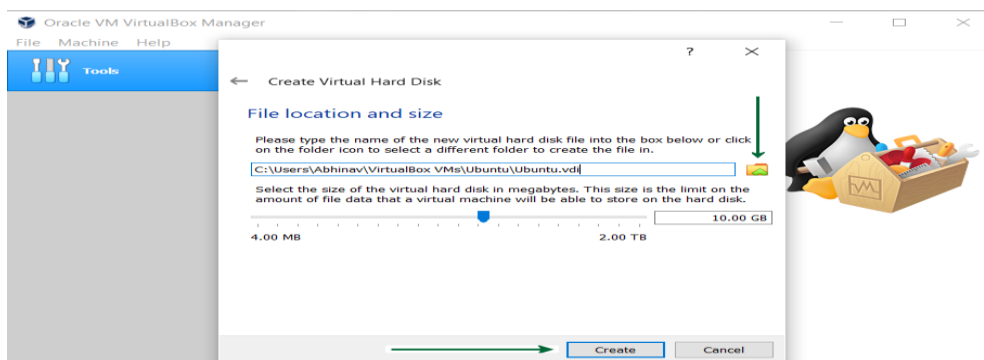
Using **VDI** type is recommended.



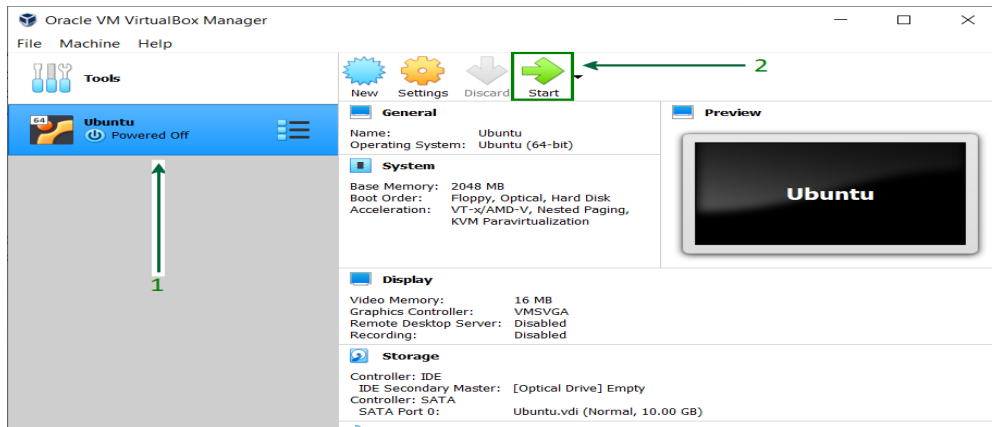
Step 7: Either of the **Physical Storage** types can be selected. Using a **Dynamically Allocated Disk** is by default recommended.



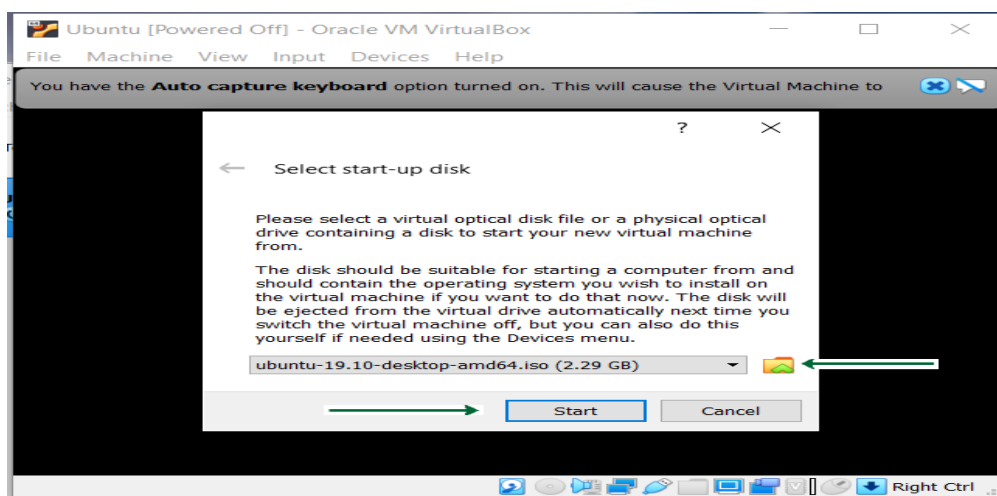
Step 8: Select **Disk Size** and provide the **Destination Folder** to install.



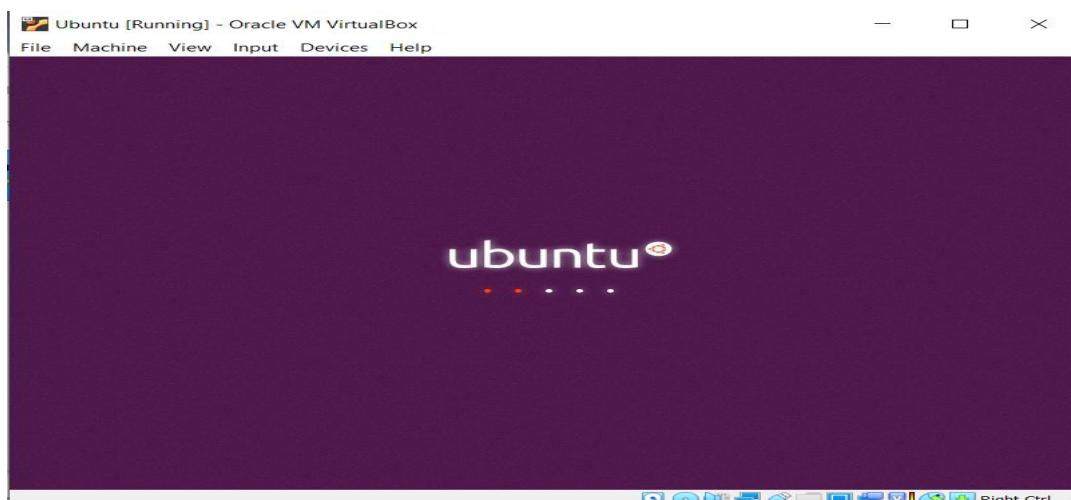
Step 9: After the Disk creation is done, boot the **Virtual Machine** and begin installing **Ubuntu**.



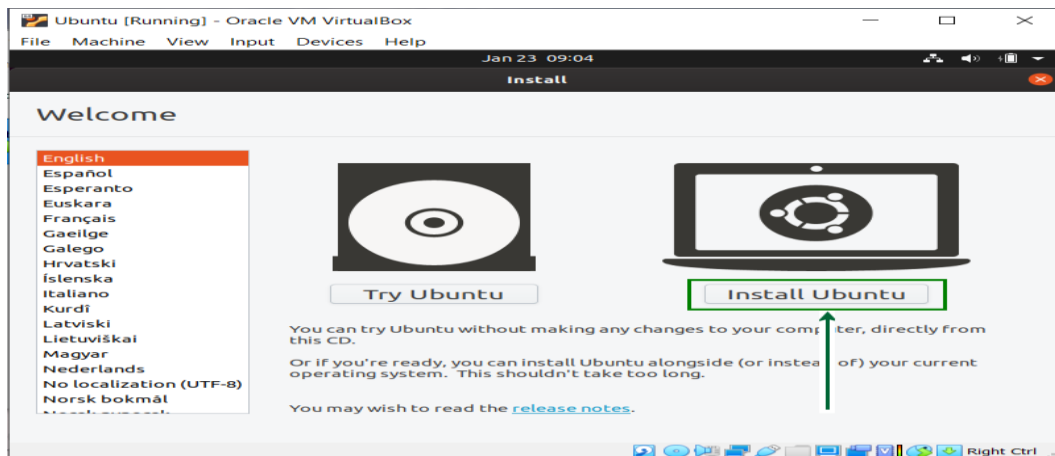
Step 10: If the installation disk is not automatically detected. Browse the file location and select the ISO file for Ubuntu.



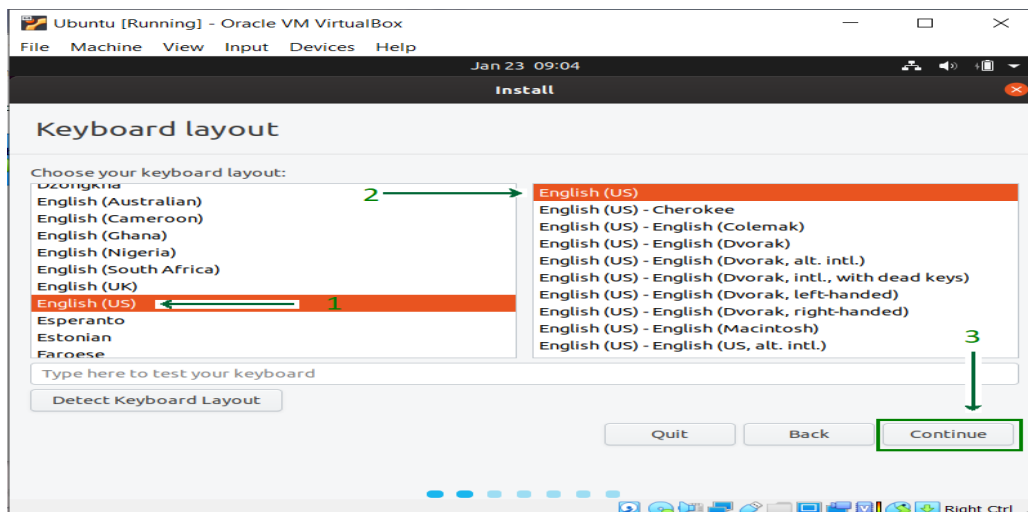
Step 11: Proceed with the installation file and wait for further options.



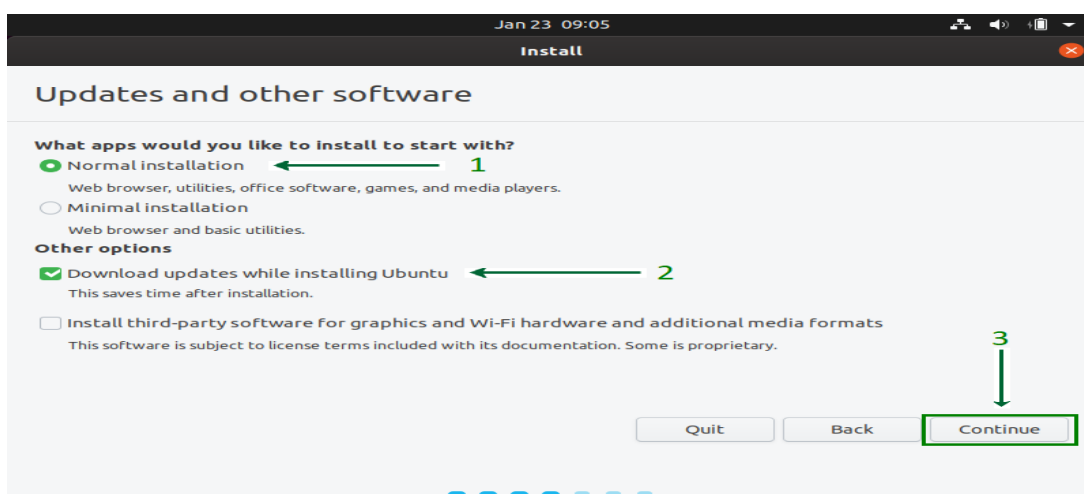
Step 12: Click on the **Install Ubuntu** option, this might look different for other Ubuntu versions.



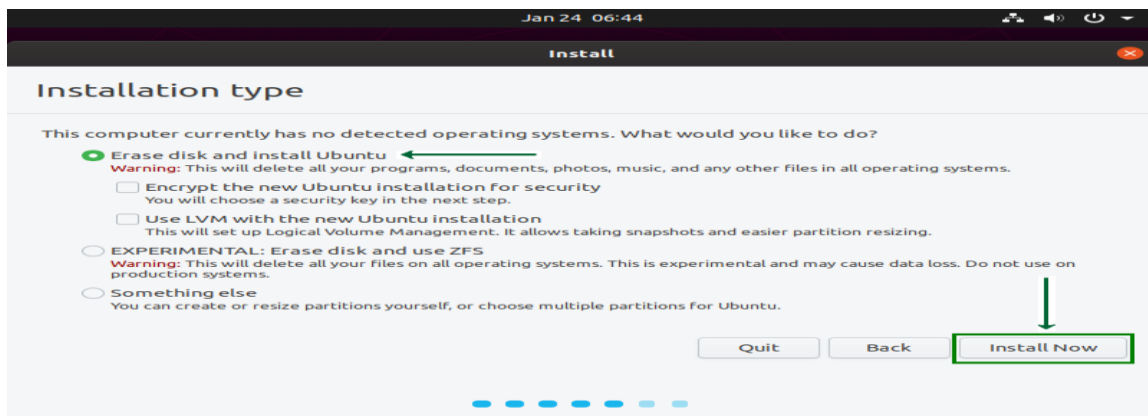
Step 13: Select **Keyboard Layout**, if the defaults are compatible, just click on the **Continue** button and proceed.



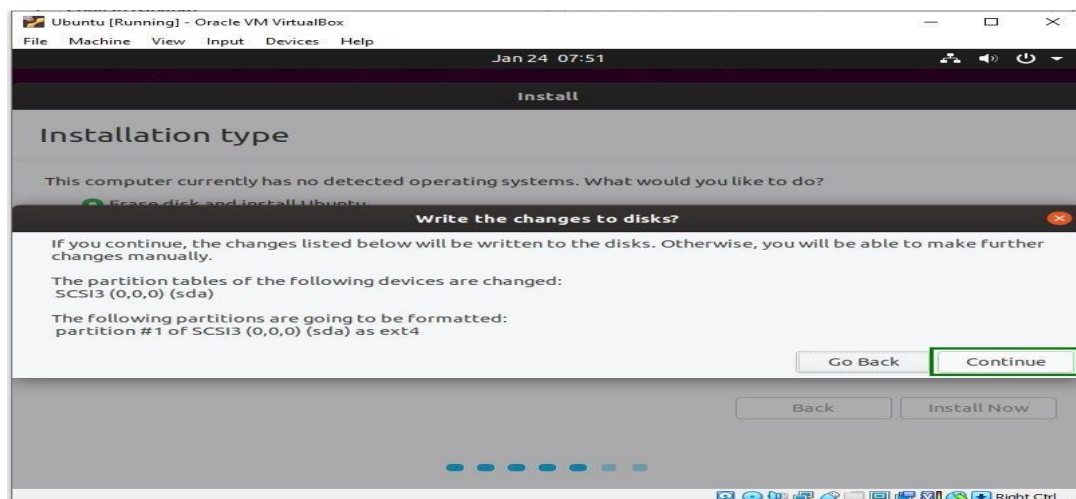
Step 14: Select **Installation Type**. By default, it is set to **Normal Installation**, which is recommended, but it can also be changed to **Minimal Installation** if there is no need for all Ubuntu features.



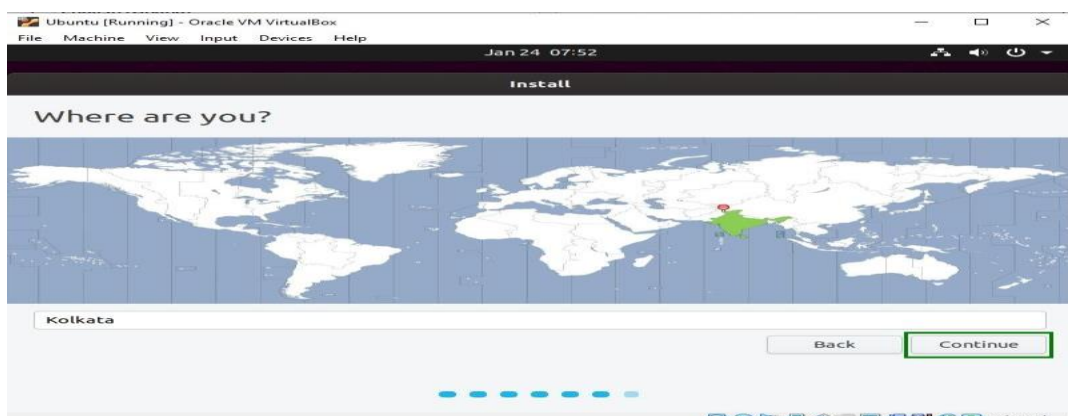
Step 15: Click on the **Install Now** button and carry on with the installation. Do not get worried about the **Erase disk** option, it will only be effective inside the virtual machine, and other system files outside the VirtualBox remain intact.



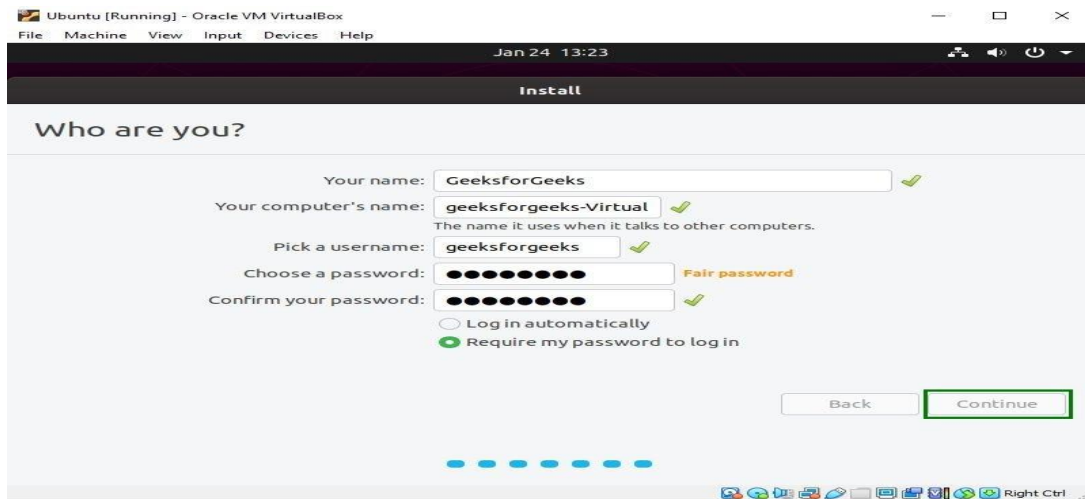
Step 16: Click on the **Continue** button, and proceed with writing changes on the disk.



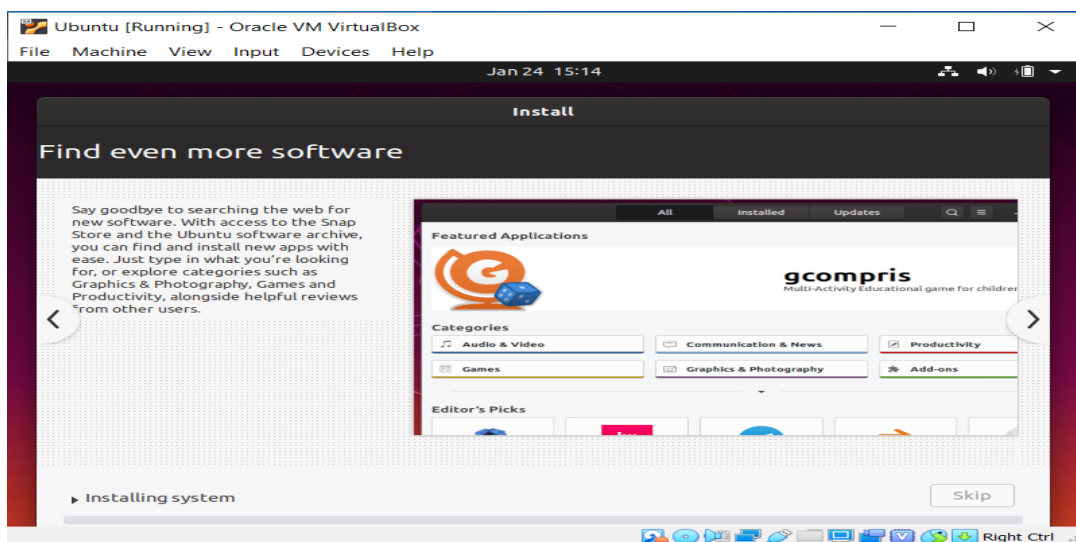
Step 17: Select your Location to set the **Time Zone**.



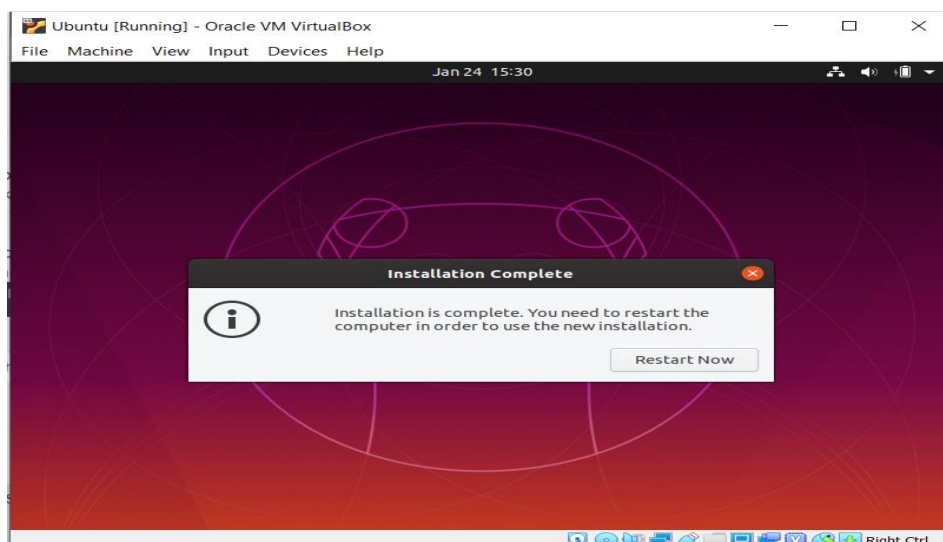
Step 18: Choose a **Name** for your computer and set a **Password** to secure login info.



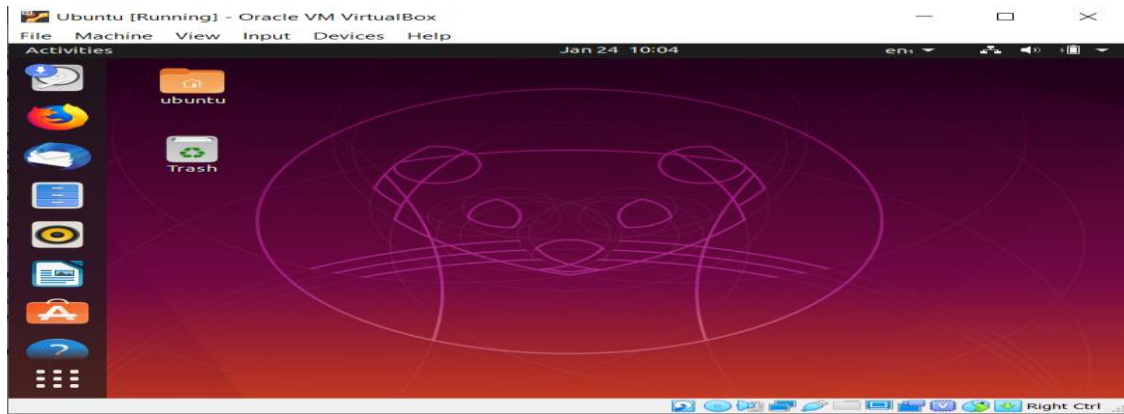
Step 19: Wait for the installation process to complete.



Step 20: Once the installation process is over, reboot your Virtual Machine.



Step 21: Voila!! You're finished with the installation process. Now you can use Ubuntu along with Windows, without creating a dual boot.



Result

Oracle VirtualBox was successfully installed on the host system. Using VirtualBox, a new virtual machine was created and Ubuntu Linux was installed without any errors. The Ubuntu operating system booted correctly and all basic functionalities such as graphical user interface, terminal access, file system operations, and network connectivity were verified. The virtual environment allowed Ubuntu to run smoothly alongside the host operating system, demonstrating effective virtualization.

Conclusion

The installation of Oracle VirtualBox and Ubuntu was completed successfully. This experiment demonstrated how virtualization technology enables multiple operating systems to run on a single physical machine efficiently. Oracle VirtualBox proved to be a reliable and user-friendly virtualization tool, while Ubuntu provided a stable and secure Linux environment. The setup is useful for learning Linux, software development, testing, and academic purposes without affecting the host system. Overall, the objective of installing and running Ubuntu using VirtualBox was achieved successfully.